

# DG Interns Hub – Cybersecurity Internship



## WEEK 4 ASSIGNMENT REPORT: TECHNICAL REPORT

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### Program Details:

**Start Date:** September 1, 2026

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**Duration:** 3 Months

**Location:** Remote

**Stipend:** Unpaid

## TABLE OF CONTENT

| PAGE | TASK | CONTENT                      |
|------|------|------------------------------|
| 1    | -    | OBJECTIVES                   |
| 2    | -    | NETWORK SECURITY             |
| 3    | 1    | Setup Basic Network (Lab)    |
| 6    | 2    | Install & Configure Services |
| 7    | 3    | Network Scanning (Nmap)      |
| 8    | 4    | Traffic Analysis (Wireshark) |

## OBJECTIVE

Understand basic network setup and perform initial security testing in a lab environment.

The goal for Week 4 was to stand up a minimal two-machine network, confirm the machines could talk to each other, and then run a first, non-destructive pass of security checks against the target: a reachability test, a service/version scan, and a look at the raw traffic generated along the way.

## LIMITED CONFIGURATION

Work was limited to configuration and observation only. That covered setting up network adapters on both virtual machines, confirming they could reach each other and that a web service on the target responded correctly, running an Nmapscan for open ports and service versions, and capturing ICMP, HTTP and DNS traffic in Wireshark. At no stage was any exploit attempted or any unauthorized access performed.

## TOOLS USED

- **VirtualBox** - Used to create and run the isolated cybersecurity lab environment.
- **Kali Linux** - Used as the attacker/security testing machine.
- **Windows 7** - Used as the target machine for hosting services and security testing.
- **Apache Web Server** - Used to host a basic web service on the target machine.
- **Nmap** - Used for network scanning, port discovery, and service identification.
- **Wireshark** - Used to capture and analyze network traffic such as HTTP, DNS, and ICMP.

# NETWORK SECURITY

Network security is the practice of protecting computer networks, devices, systems, and data from unauthorized access, attacks, misuse, modification, or disruption.

A host becomes risky the moment a service on it is reachable by more people (or machines) than intended. An outdated file-transfer daemon, an admin panel left open, or a database port facing the wrong network segment can each turn into the first foothold an attacker needs. Because of this, keeping an accurate, current inventory of what a network actually exposes is one of the most basic and most valuable security habits an organization can build.

**Confidentiality** - Only authorized users can access information.

**Integrity** - Data should not be altered without authorization.

**Availability** - Network services and resources should remain available when needed.

## What does the scanning tell us

It tells us about when the machines are alive, which doors (ports) are unlocked on them, and what is standing behind each door. This information is exactly what an attacker would gather before attempting anything further, which is why defenders run the same checks themselves — to close gaps before someone else finds them.

## Reading Traffic with a Packet Analyzer

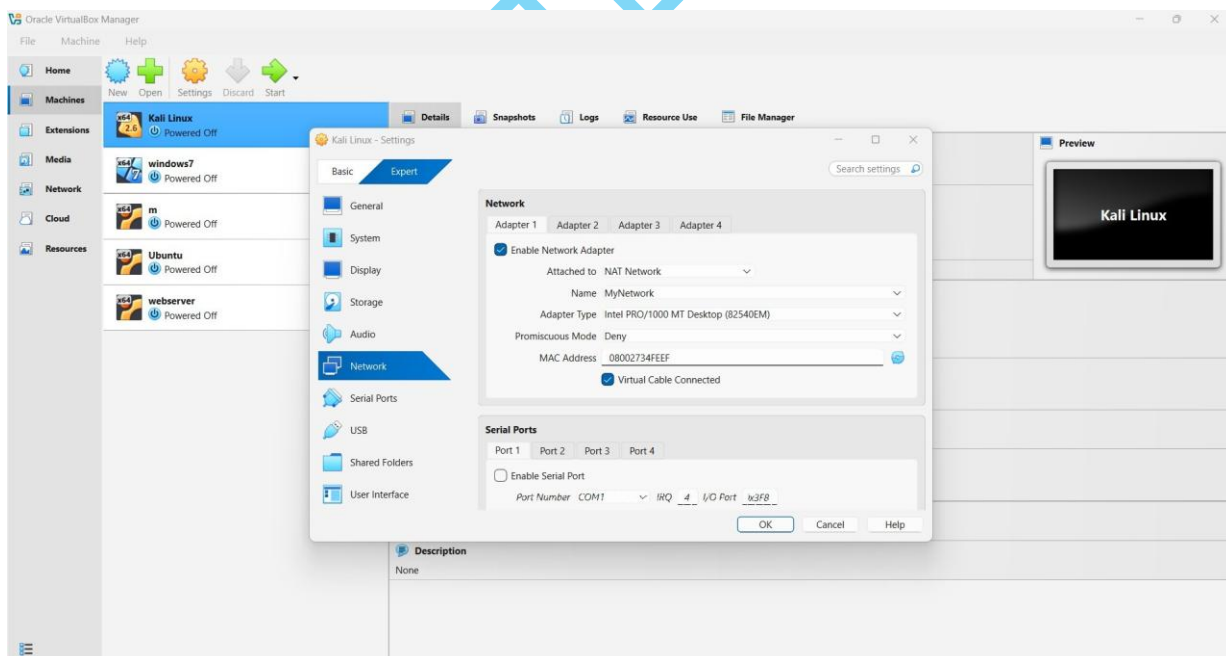
Scanning shows what is exposed; packet capture shows how that exposure actually behaves on the wire. Tools such as Wireshark record every frame crossing an interface, letting an analyst confirm whether a protocol like HTTP, DNS or ICMP is functioning as expected, and — just as importantly — whether the data inside it is protected or sitting in plain view.

# TASK - 1 Setup Basic Network

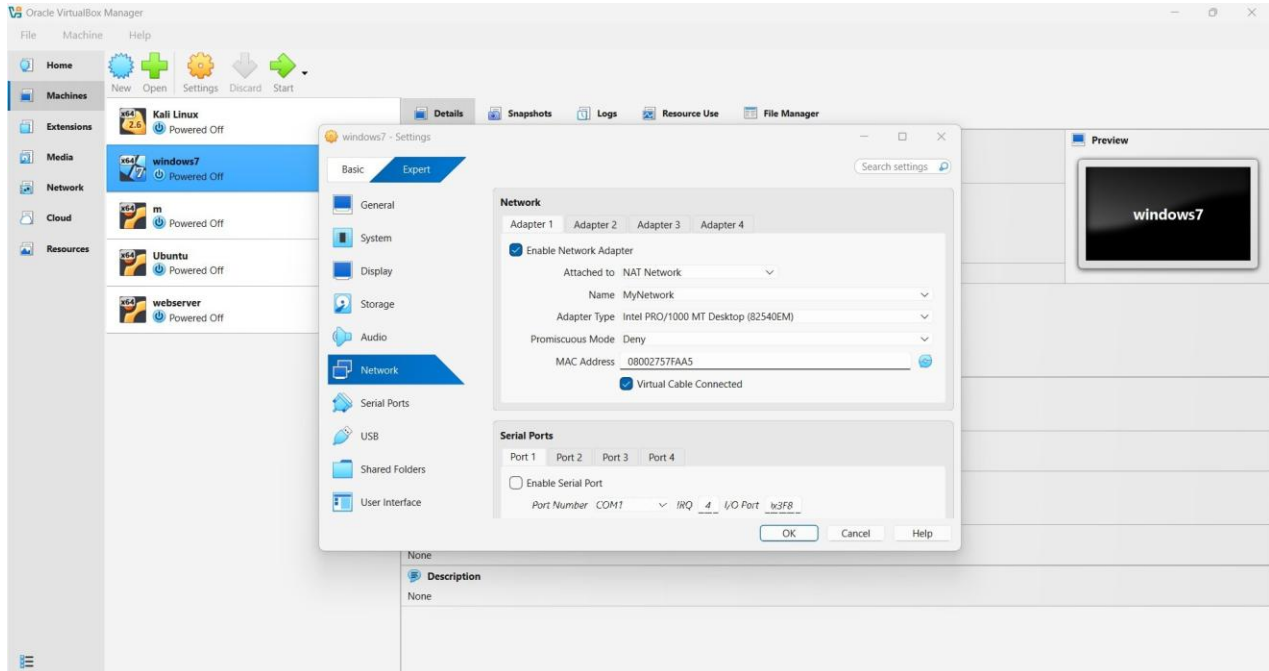
The purpose of **Task 1** is to create a small, isolated network environment for performing cybersecurity testing safely. The lab contains at least two virtual machines connected to the same VirtualBox network.

In this project:

- **Kali Linux (10.0.2.4)** is used as the **attacker/security testing machine**.
- **Windows 7 (10.0.2.3)** is used as the **target machine**.
- Both machines are connected to the **10.0.2.0/24 network**.
- The network connection allows Kali to communicate with the Windows 7 target for authorized testing.
- **Kali Linux** and **Windows 7** have the same network **NAT Network**.
- **Kali Linux :**

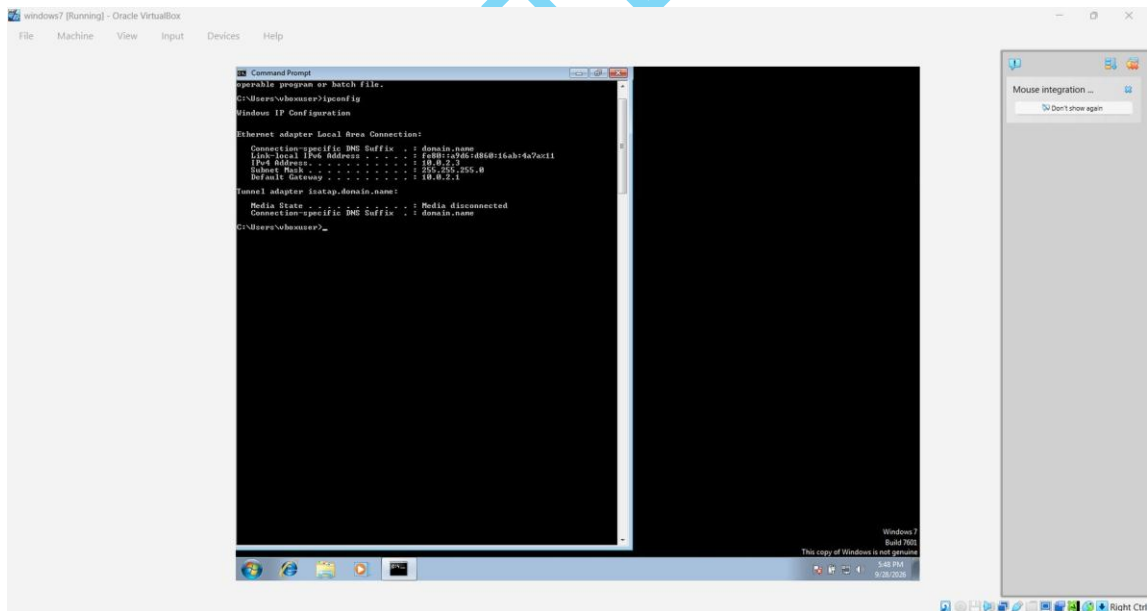


## Windows 7 :



## Connection Checks :

Check the ip address of Windows 7 in the cmd of Windows 7 and Ping Windows 7 ip address in the Kali Linux.



Session Actions Edit View Help

(kali@kali)-[~]

\$ ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:34:fe:ef brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.4/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 340sec preferred_lft 340sec
    inet6 fe80::541c:c6cd:25a1:a94a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

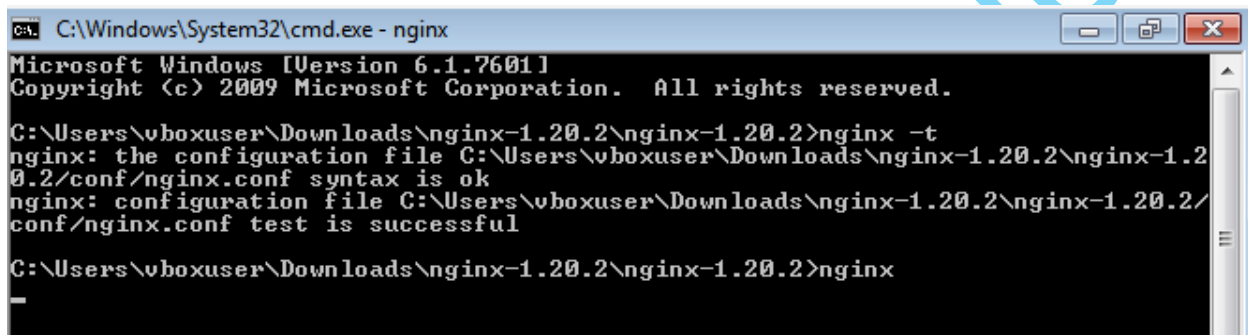
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## Task 2: Install & Configure Services

### Web Server Configuration:

Nginx version 1.20.2 was installed and configured on the Windows 7 target machine. The Nginx configuration was validated successfully using

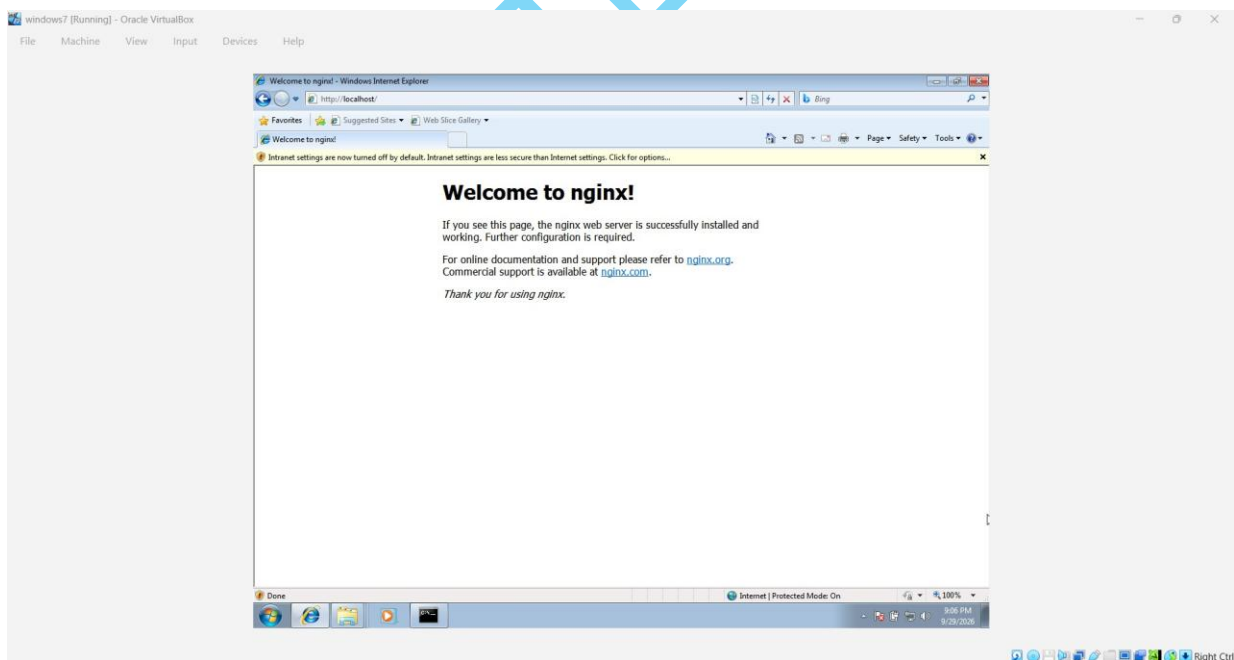
`nginx -t`. The server was then started and verified through a web browser by accessing `http://localhost`. The default Nginx welcome page was displayed, confirming that the web server was running successfully on TCP port 80.



```
C:\Windows\System32\cmd.exe - nginx
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\vboxuser\Downloads\nginx-1.20.2\nginx-1.20.2>nginx -t
nginx: the configuration file C:\Users\vboxuser\Downloads\nginx-1.20.2\nginx-1.20.2\conf/nginx.conf syntax is ok
nginx: configuration file C:\Users\vboxuser\Downloads\nginx-1.20.2\nginx-1.20.2\conf/nginx.conf test is successful

C:\Users\vboxuser\Downloads\nginx-1.20.2\nginx-1.20.2>nginx
```



## TASK 3: Network Scanning (Nmap)

Nmap service-version detection was performed against the Windows 7 target at 10.0.2.3. The host was found to be active. TCP port 80 was identified as open and running HTTP with Nginx version 1.20.2. The remaining 999 scanned TCP ports were reported as filtered, indicating that they did not respond to the scan. The scan confirmed that the configured web service was accessible from the Kali Linux testing machine.

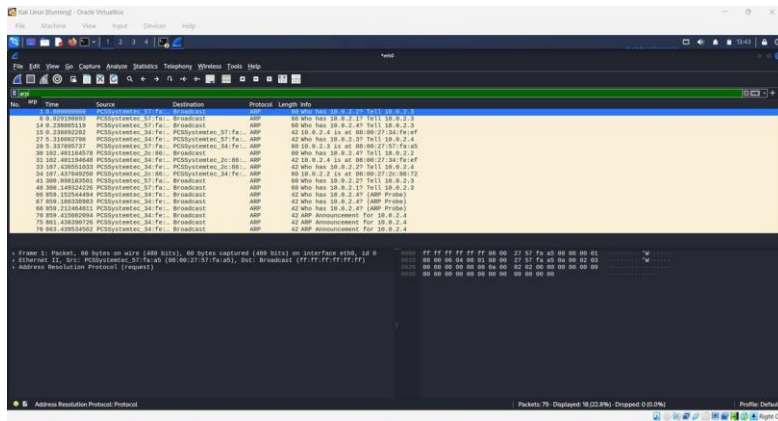
```
(kali㉿kali)-[~]
$ nmap -sV 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-29 11:45 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00086s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http   nginx 1.20.2
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.90 seconds
```

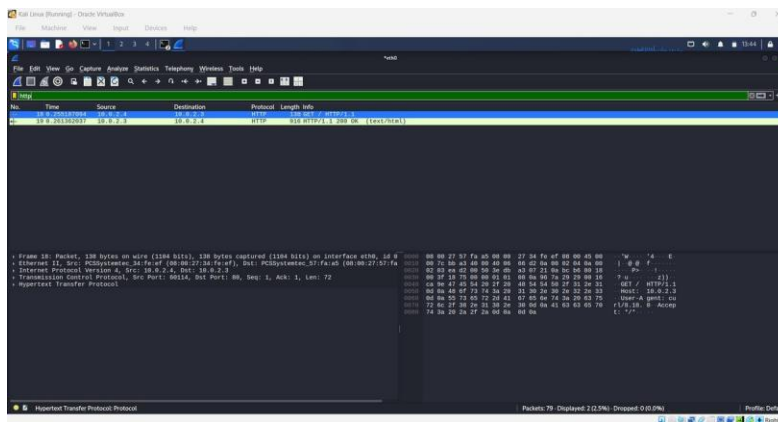
| Scan Detail         |          | Information             |       | Value / Output                                    |              |
|---------------------|----------|-------------------------|-------|---------------------------------------------------|--------------|
| Target IP           |          | Scanned host IP address |       | 10.0.2.3                                          |              |
| Scan Type           |          | Nmap scan type used     |       | -sV (Service Version Detection)                   |              |
| Nmap Version        |          | Version of Nmap used    |       | 7.98                                              |              |
| Start Time          |          | Scan start time         |       | 2026-09-29 11:45 -0400                            |              |
| Host Status         |          | Host is up/latency      |       | Host is up (0.00086s latency)                     |              |
| Total Ports Scanned |          | Total TCP ports scanned |       | 1000 (1 open, 999 filtered)                       |              |
| Open Ports Detected |          |                         |       |                                                   |              |
| Port                | Protocol |                         | State | Service                                           | Version      |
| 80                  | TCP      |                         | Open  | HTTP                                              | nginx 1.20.2 |
| MAC Address         |          | Target MAC address      |       | 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC) |              |

# TASK 4: Traffic Analysis (Wireshark)

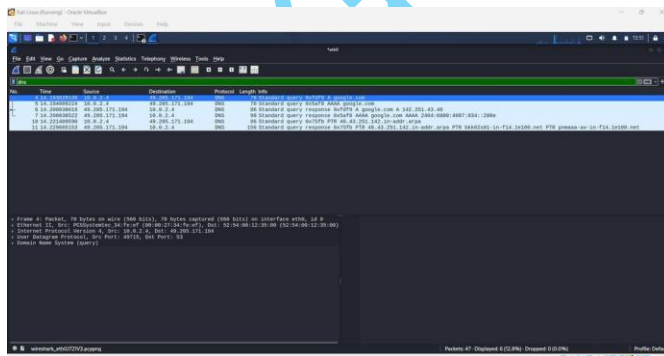
## ARP :



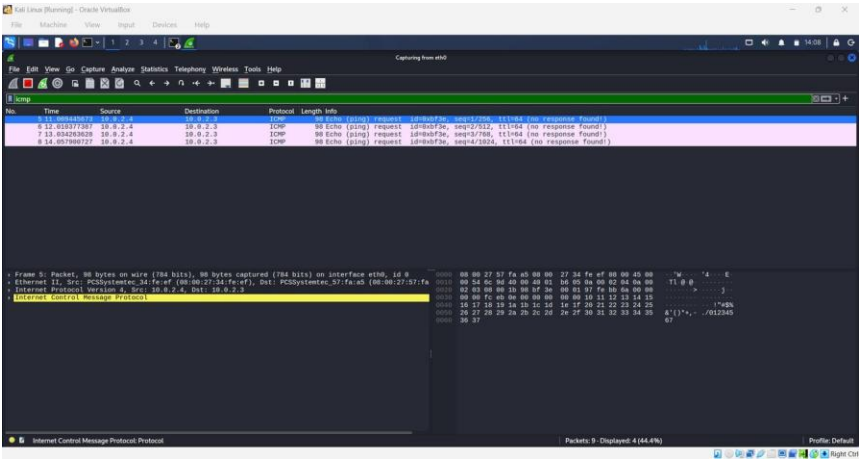
## HTTP :



## DNS :



ICMP :



MODERN CYBERSECURITY DASHBOARD: MULTI-CAPTURE AUDIT

| PROTOCOL CAPTURE                                                                                     | SECURITY CHECK CATEGORY                  | KEY OBSERVATIONS                                                                                                   | RISK LEVEL |
|------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------|
| <br>IMAGE 1: ARP    | Network Layer<br>Spoofing & Recon        | Multiple gratuitous ARP announcements from specific IPs, suggesting potential port scanning or poisoning attempts. | MEDIUM     |
| <br>IMAGE 2: HTTP  | Data Privacy &<br>Cleartext Transmission | Plaintext GET request and User-Agent ('curl/8.18.0') visible. HTTP 200 payload data accessible without encryption. | HIGH       |
| <br>IMAGE 3: DNS  | Domain Resolution<br>Integrity           | Unencrypted A, AAAA, and Reverse PTR lookups to an external DNS resolver (49.205.171.194), vulnerable to snooping. | LOW-MEDIUM |
| <br>IMAGE 4: ICMP | Host Discovery &<br>Availability         | Repeated ICMP Echo Requests (Pings) sent from 10.0.2.4 to 10.0.2.3 with zero responses recorded.                   | LOW        |

## Task 5: Mini Project Output

- I set up a bridged virtual lab network from scratch
- I verified host-to-host reachability with ping and curl
- I found open ports and running services using Nmap
- Read and filtered live traffic in Wireshark
- Recognized everyday risks like cleartext protocols
- I understood why testing must stay inside an authorized lab
- It also reinforced a simple rule for any future testing:  
scope everything to a lab that has been explicitly built and  
authorized for it.

**THANK YOU**

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